

layout: post title: MiniPrep Plus DNA/RNA extractions date: '2024-04-03' categories: tags: [DNA, RNA, Extractions, Protocols]

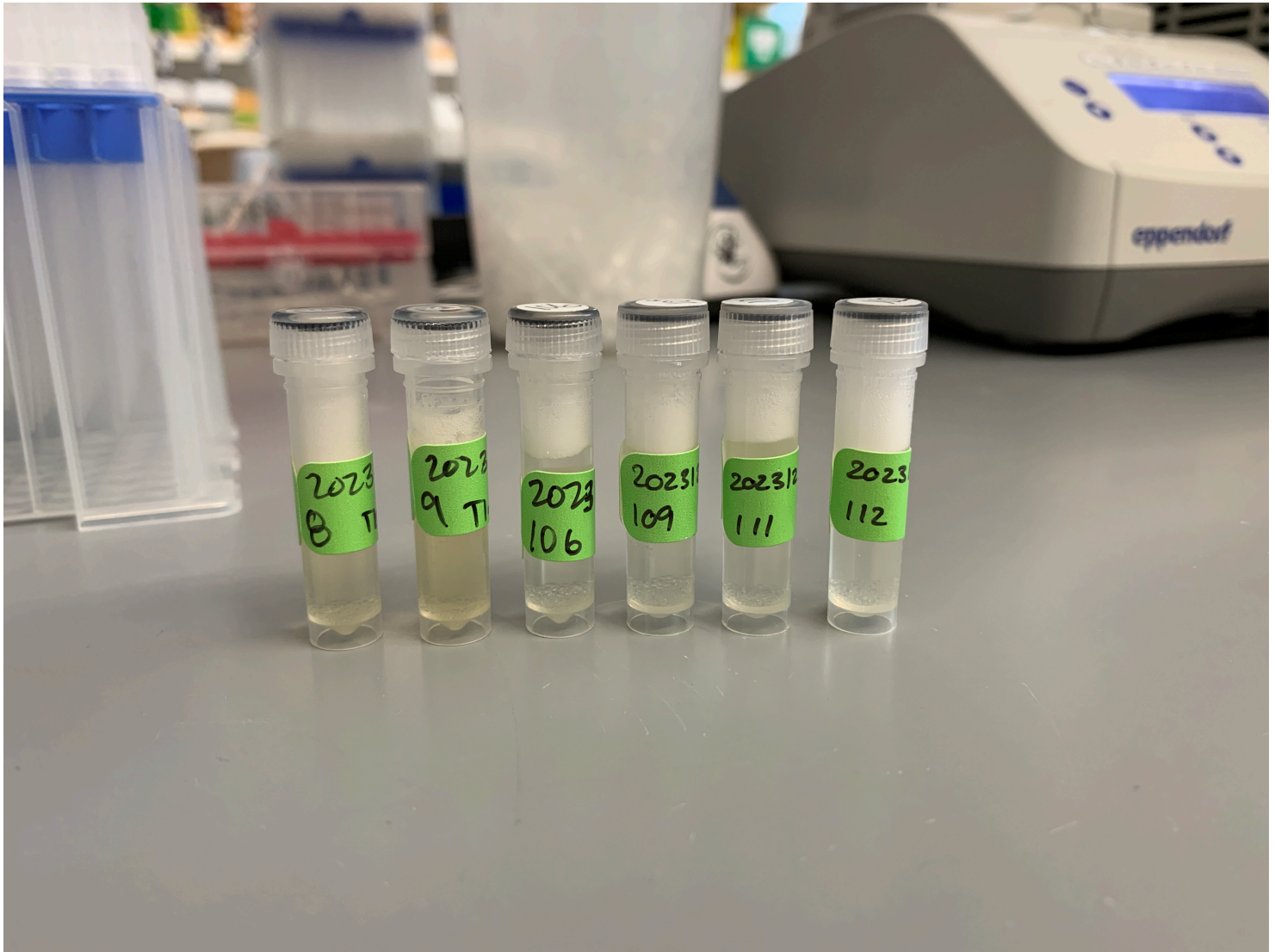
projects: Moorea Poc spawning 2023

Extractions for Moorea Pocillopora spawning 2023

This protocol is based on Maggie's coral egg/larvae extraction [protocol](#) and Kevin's coral larvae extraction [protocol](#). This post details the info about the extraction steps for larval *Pocillopora* samples from Moorea 2023. The github for that project is linked [here](#).

Samples

The samples run today were 8, 9, 106, 109, 111, and 112. These samples had ~400 ul of larvae per tube and were preserved in 500 uL of DNA/RNA Shield. They were transported in a freezer box that also included formalin-fixed samples and were stored at 4°C for ~3 months. This photo is post-bead beating:



Materials

- Zymo Quick-DNA/RNA Miniprep Plus Kit [HERE](#) Protocol Booklet
- Tris-Ethylenediaminetetraacetic acid (EDTA) 1X buffer for DNA elution
- Heating block capable of heating to 70°C
- Centrifuge and rotor capable of spinning at 15,000 rcf
- Plastics
 - 3 1.5 mL microcentrifuge tubes per sample
 - 2 PCR tubes per sample
 - 2 Qubit tubes per sample
 - 1 5 mL tube per sample

Protocol

Protocol was followed according to this [post](#). A few differences:

- I added 500 uL of DNA/RNA shield to the samples after thawing them so that the volume in the tubes was ~1 mL of shield. I aliquoted out 500 uL for extractions and saved the remaining fraction in case we need to re-extract.
- In the RNA extraction, I added another wash step with 400 uL of wash buffer after the DNase incubation (i.e., after the incubation, I did 400 uL of prep buffer, 700 uL of wash buffer, 400 uL of wash buffer, and another 400 uL of wash buffer).
- I eluted in 80 uL of either Tris (DNA) or DNA/RNA free water (RNA) instead of 100 uL.
- In the second extraction of the day, I tipped the RNA samples onto the floor when trying to close lids to the caps. This was during the second incubation for RNA elution.

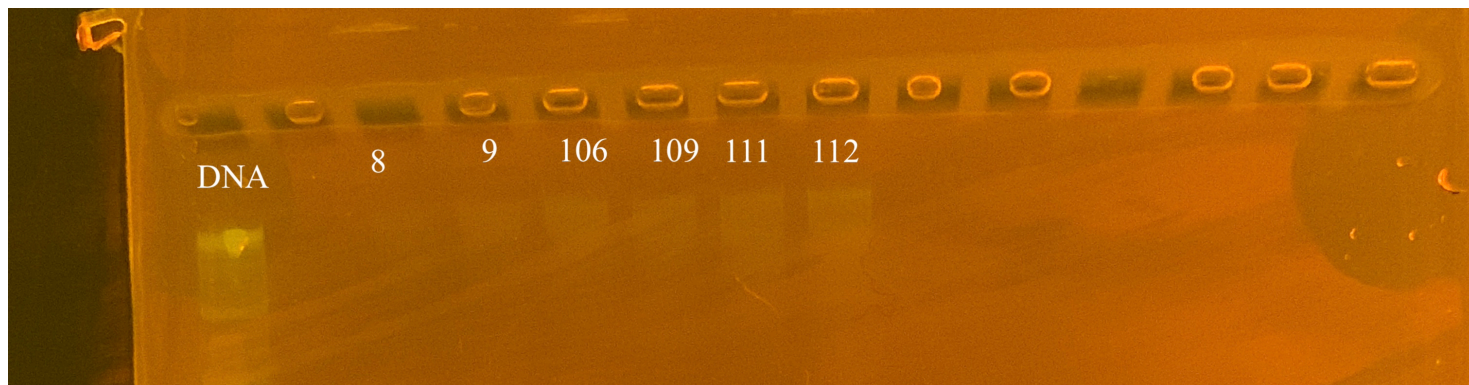
QC

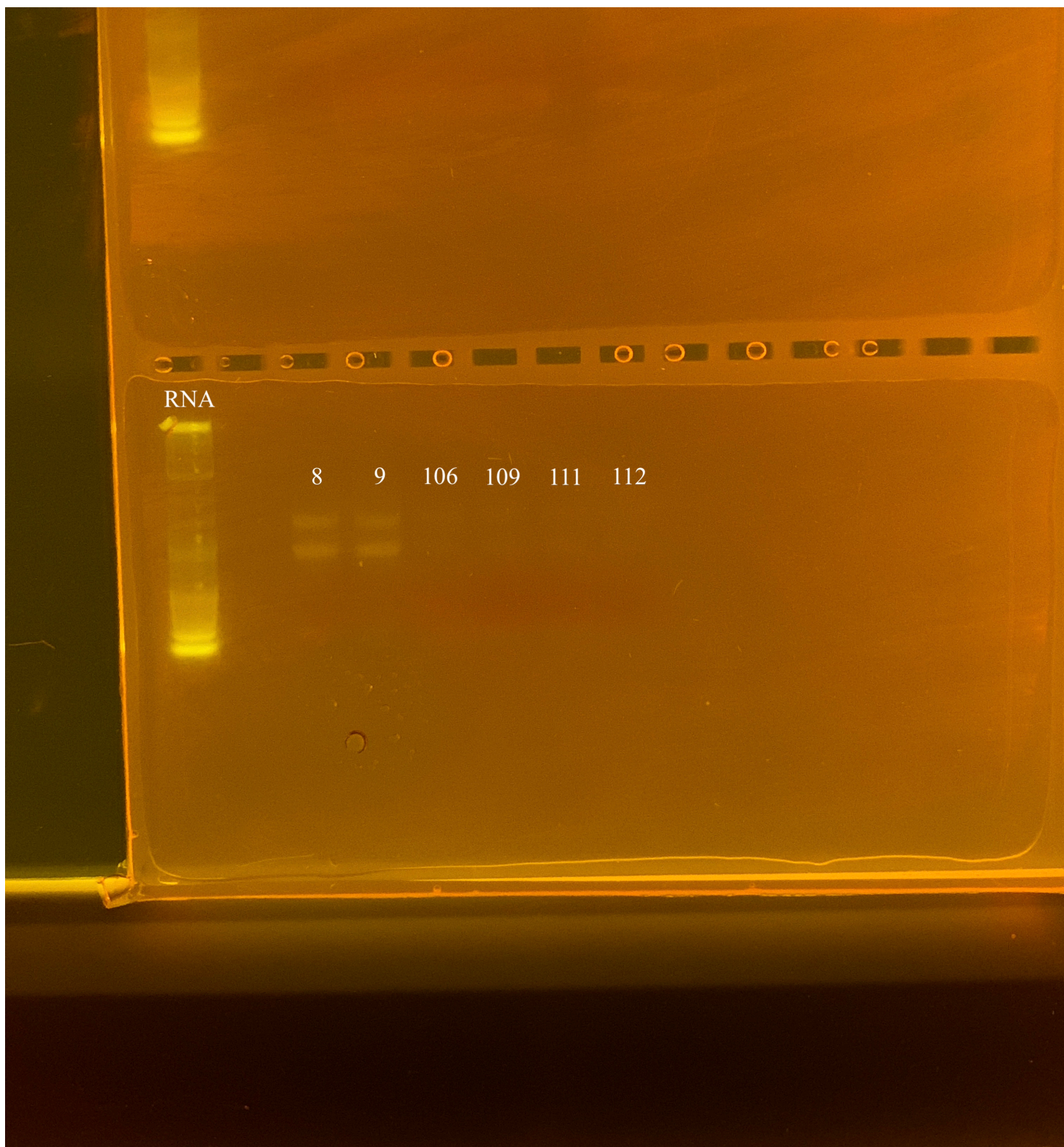
All QC was run the same day as extractions.

Qubit results

Tube.ID	DNA1 <i>nguL</i>	DNA2 <i>nguL</i>	DNAavgng_uL	RNA1 <i>nguL</i>	RNA2 <i>nguL</i>	RNAavgng_uL
8	NA	NA	NA	29.6	29.4	29.5
9	NA	NA	NA	46	46.4	46.2
106	3.48	3.46	3.47	11	11	11
109	4.54	4.56	4.55	10.4	10.2	10.3
111	3.96	4	3.98	NA	NA	NA
112	3.52	3.52	3.52	NA	NA	NA

Gel



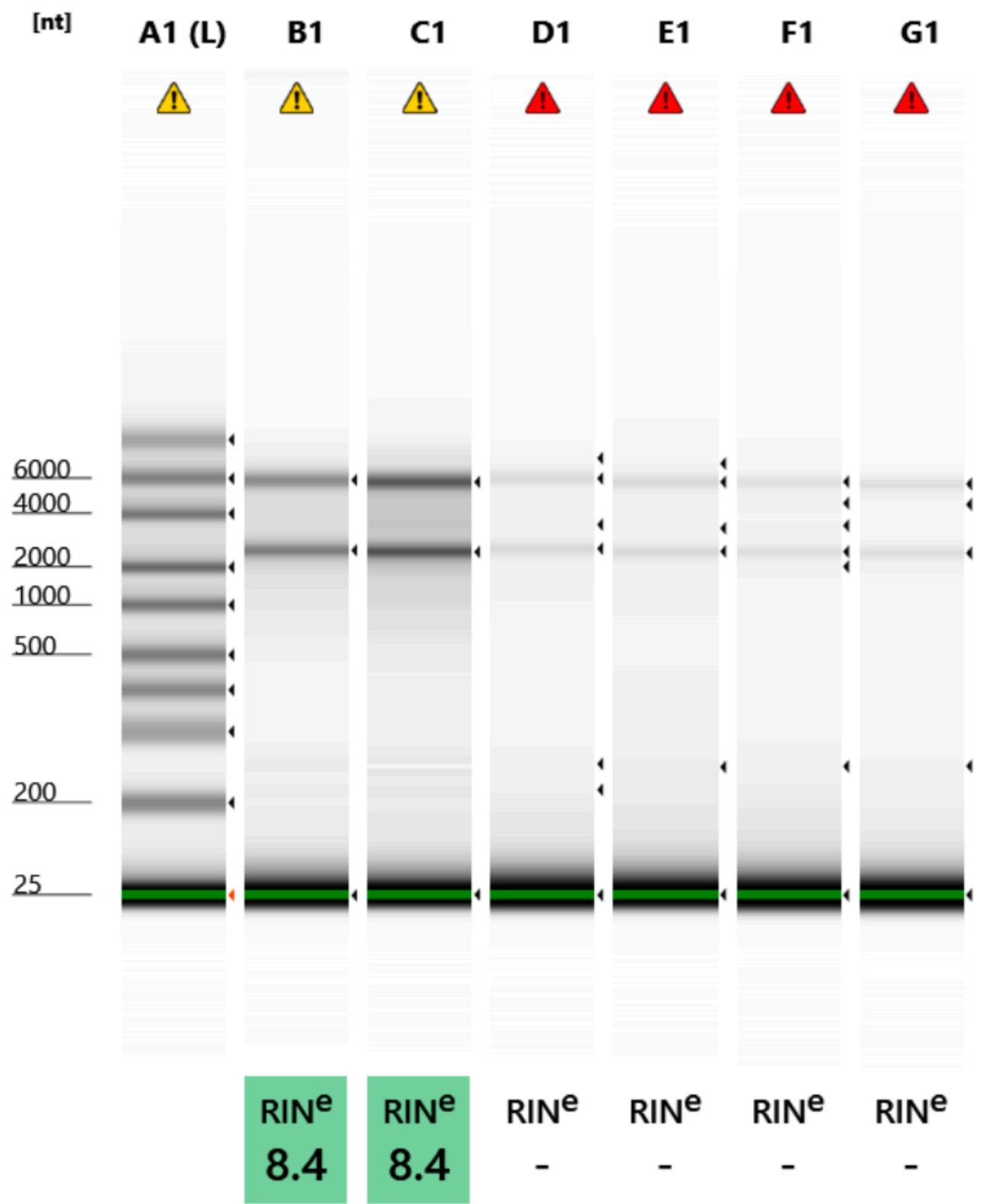


There are very very very faint bands in the RNA samples for the last 4 samples. I did not get Qubit values for the last 2 samples but hopefully will get tapestation concentrations for them.

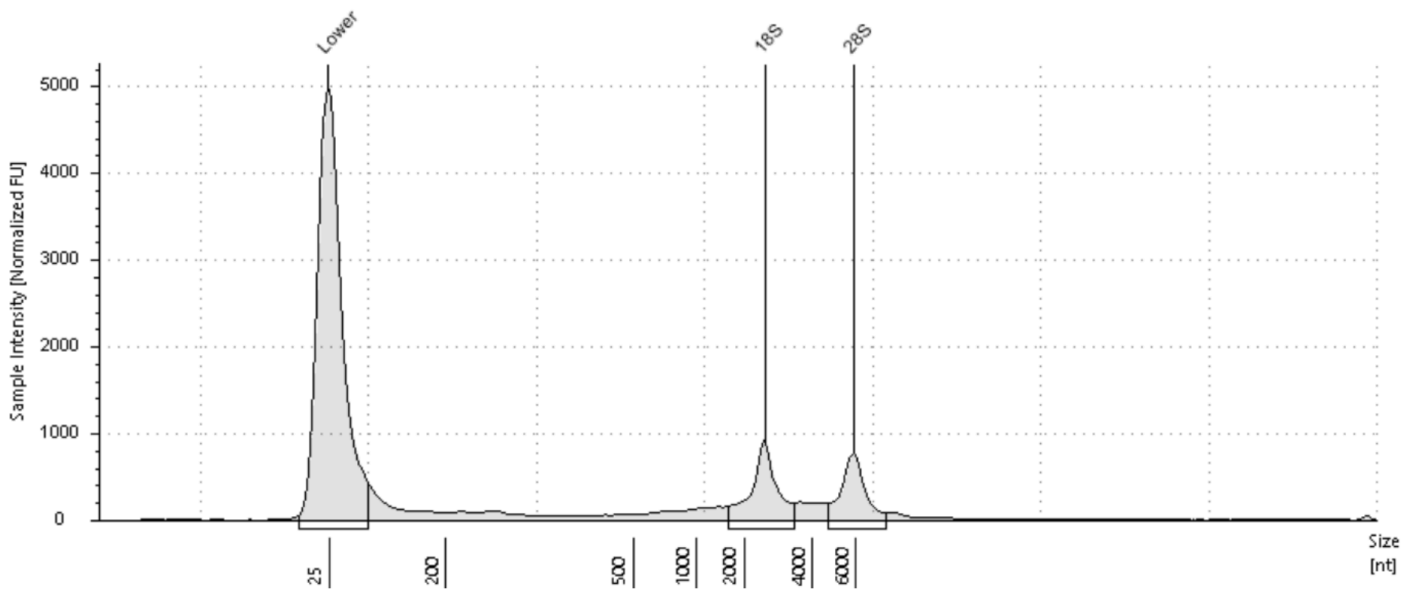
Tapestation

See full tapestation report [here](#).

Filename: 2024-04-03 - 14.02.26.RNA



B1: 8



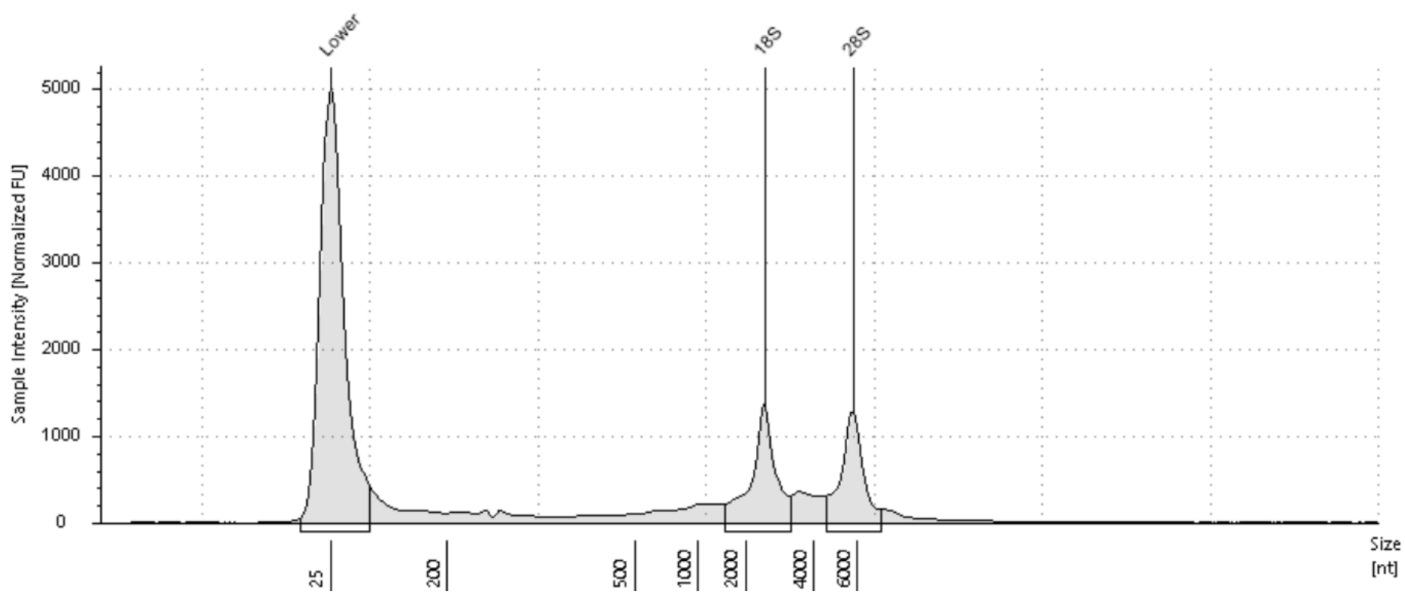
Sample Table

Well	RINe	28S/18S (Area)	Conc. [ng/μl]	Sample Description	Alert	Observations
B1	8.4	0.8	15.9	8		Caution! Expired ScreenTape device; Sample concentration outside functional range for RINe and the assay

Peak Table

Size [nt]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
25	40.0	40.0	4710	-		Lower Marker
2504	4.42	-	5.19	55.30		18S
5934	3.57	-	1.77	44.70		28S

C1: 9



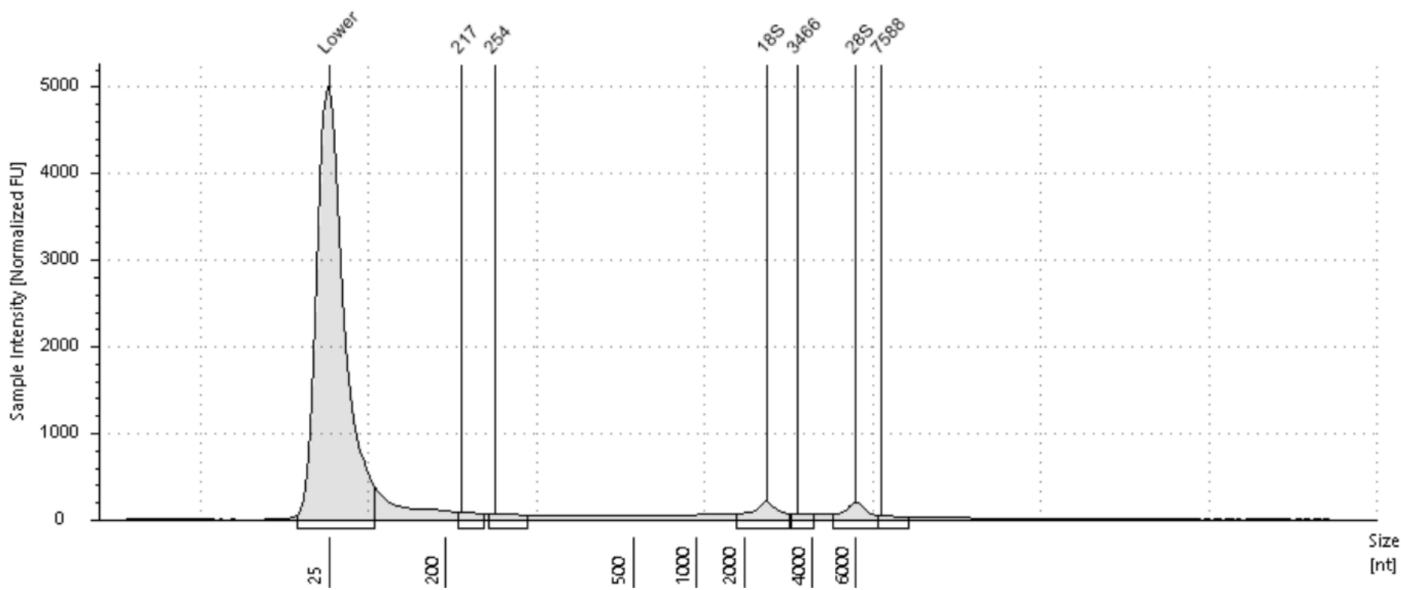
Sample Table

Well	RINe	28S/18S (Area)	Conc. [ng/μl]	Sample Description	Alert	Observations
C1	8.4	0.9	23.2	9		Caution! Expired ScreenTape device; Sample concentration outside functional range for RINe and the assay

Peak Table

Size [nt]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
25	40.0	40.0	4710	-		Lower Marker
2429	6.44	-	7.80	53.73		18S
5755	5.55	-	2.84	46.27		28S

D1: 106



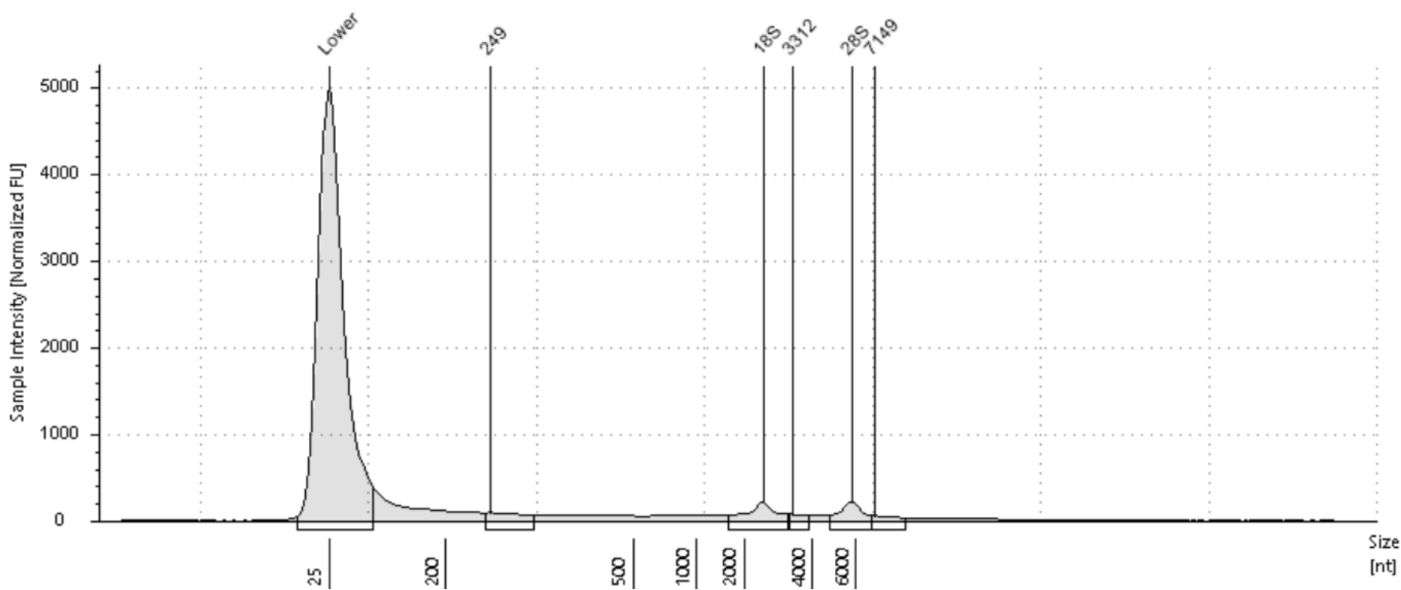
Sample Table

Well	RINe	28S/18S (Area)	Conc. [ng/μl]	Sample Description	Alert	Observations
D1	-	-	7.29	106		Caution! Expired ScreenTape device; Sample concentration outside functional range for RINe and the assay

Peak Table

Size [nt]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
25	40.0	40.0	4710	-		Lower Marker
217	0.330	-	4.47	11.33		
254	0.386	-	4.47	13.27		
2540	0.963	-	1.11	33.08		18S
3466	0.247	-	0.210	8.50		
6000	0.799	-	0.392	27.47		28S
7588	0.185	-	0.0715	6.34		

E1: 109



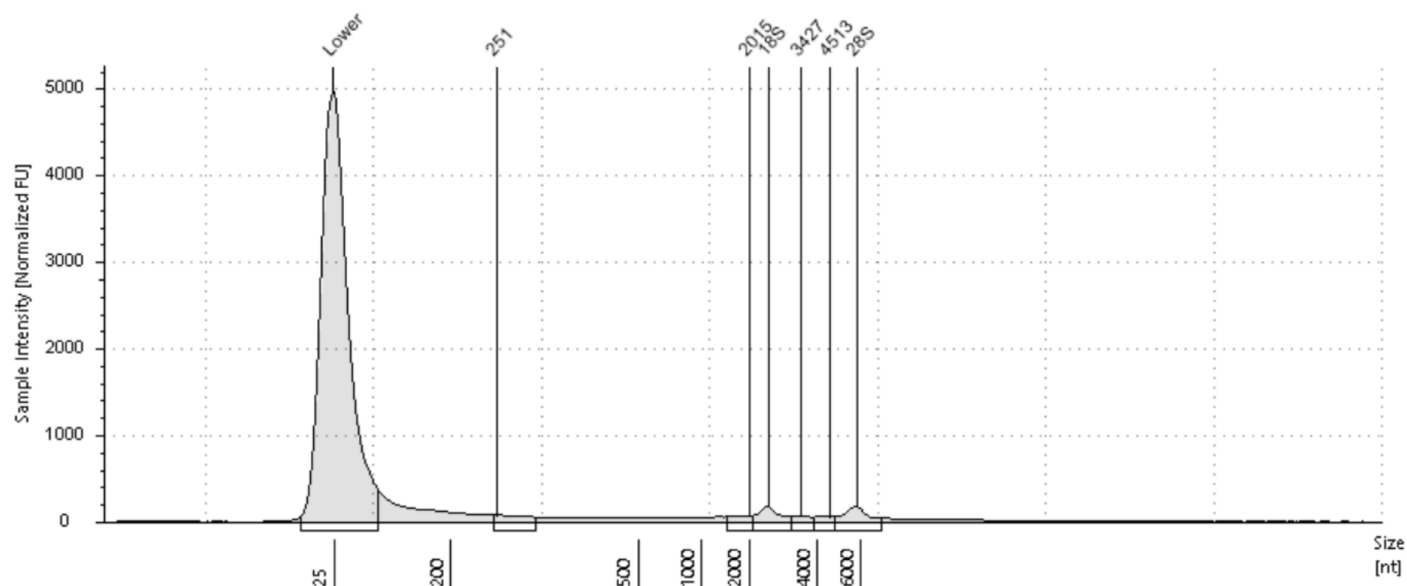
Sample Table

Well	RINe	28S/18S (Area)	Conc. [ng/μl]	Sample Description	Alert	Observations
E1	-	-	8.71	109		Caution! Expired ScreenTape device; Sample concentration outside functional range for RINe and the assay

Peak Table

Size [nt]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
25	40.0	40.0	4710	-		Lower Marker
249	0.637	-	7.52	20.63		
2455	1.08	-	1.30	35.10		18S
3312	0.231	-	0.205	7.48		
5772	0.873	-	0.445	28.26		28S
7149	0.263	-	0.108	8.53		

F1: 111



Sample Table

Well	RINe	28S/18S (Area)	Conc. [ng/μl]	Sample Description	Alert	Observations
F1	-	-	7.23	111	▲	Caution! Expired ScreenTape device; Sample concentration outside functional range for RINe and the assay

Peak Table

Size [nt]	Calibrated Conc. [ng/μl]	Assigned Conc. [ng/μl]	Peak Molarity [nmol/l]	% Integrated Area	Peak Comment	Observations
25	40.0	40.0	4710	-		Lower Marker
251	0.454	-	5.33	18.31		
2015	0.267	-	0.390	10.76		
2444	0.649	-	0.780	26.13		18S
3427	0.217	-	0.186	8.74		
4513	0.207	-	0.135	8.33		
5783	0.688	-	0.350	27.73		28S

Tapestation looks great! Concentrations are all relatively high, even in the samples that Qubit did not pick up. There are also two bands in the TS gel. These look good to go for library prep.